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DECODELABS ADVANCED TRAINING SUITE: MULTI-MODEL ML BENCHMARK
Batch: 2026 | Powered by DecodeLabs
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[FEATURE 1] Initiating Dynamic Hyperparameter Optimization Loop...
-> Tuning Complete: Mathematically Optimal K found at K = 2
-> Minimum Test Error Achieved: 0.0000
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[FEATURE 2] Executing Comparative Multi-Model Showdown Suite...
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===== BENCHMARK SHOWDOWN MATRIX =====
Metric          | K-Nearest Neighbors (K=2) | Logistic Regression
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Accuracy Score   |          100.00           % |          100.00           %
Macro F1-Coefficient |          1.0000           |          1.0000
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[FEATURE 4] Advanced Diagnostic Layer Matrix:
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--- KNN Confusion Matrix ---
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```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

```
--- Logistic Regression Confusion Matrix ---
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
-----
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[FEATURE 3] Launching Live Deployment serving prompt...  
-> Continuous production loop active. Enter data to serve instant inference.  
-> Type 'exit' at any prompt to close the pipeline session.
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--- New Sample Inference Vector ---  
Enter Sepal Length (cm): 5.1  
Enter Sepal Width (cm): 3.5  
Enter Petal Length (cm): 1.4  
Enter Petal Width (cm): 0.2
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>>> MACHINE LEARNING DECISION VERDICT: Iris SETOSA <<<
```

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[LOGGED] Verification saved to internal file tracker: inference_history.txt
```

```
--- New Sample Inference Vector ---  
Enter Sepal Length (cm): 6.4  
Enter Sepal Width (cm): 2.9  
Enter Petal Length (cm): 4.3  
Enter Petal Width (cm): 1.3
```

```
>>> MACHINE LEARNING DECISION VERDICT: Iris VERSICOLOR <<<
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[LOGGED] Verification saved to internal file tracker: inference_history.txt
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```
--- New Sample Inference Vector ---  
Enter Sepal Length (cm): 6.9  
Enter Sepal Width (cm): 3.1  
Enter Petal Length (cm): 5.4  
Enter Petal Width (cm): 2.1
```

```
>>> MACHINE LEARNING DECISION VERDICT: Iris VIRGINICA <<<
```

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[LOGGED] Verification saved to internal file tracker: inference_history.txt
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```
--- New Sample Inference Vector ---  
Enter Sepal Length (cm): exit
```

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=====  
PIPELINE TERMINATED: PORTFOLIO VERIFICATION SUITE COMPLETE  
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prj2.py

inference_history.txt ✕

inference_history.txt

```
1 Input: [[5.1, 3.5, 1.4, 0.2]] -> Prediction: Iris SETOSA
2 Input: [[6.4, 2.9, 4.3, 1.3]] -> Prediction: Iris VERSICOLOR
3 Input: [[6.9, 3.1, 5.4, 2.1]] -> Prediction: Iris VIRGINICA
4
```